

Project Proposal

Initial Draft

I. Project Title

TransitWays: A Commute route and delay Adjuster and Assistance

II. Problem Statement

Commuters using public transportation often face unexpected delays due to traffic, accidents and many problems involving the road. Existing transit apps shows schedules but don't always inform us about possible alternatives when a specific place or line is blocked. This problem is worth solving to ensure efficient travel time to everyone as many rely on public transportation.

III. Project Objectives

- Store transit route and their baseline travel durations
- Delay times inserted into standard transit path
- Automatically suggest a new transport if there's a delay exceeding 20 minutes.

IV. Planned Features

- Route status monitor: Checks if a transpo is "on-time" or "delayed"
- Route Redirector and Assistance: Gives backup option if travel delays disrupt a commute too much.

V. Planned Inputs and Outputs

Inputs: Desired route (Route A or Route B), Incident Report Input (Accident, Traffic, None)

Outputs: Baseline travel time, calculated delay, final estimated arrival time, and recommendations of redirection.

VI. Logic Plan (Pseudocode)

START

SET route_a_base_time = 30 // minutes

SET route_b_base_time = 45 // minutes

SET delay_time = 0

PRINT "Select your regular route (A or B): "

INPUT user_route

PRINT "Enter current transit condition (Accident, Traffic, None)

INPUT condition

IF condition == "Traffic" THEN

 SET delay_time = 10

ELSE IF condition == "Accident" THEN

 SET delay_time = 25

ENDIF

IF user_route == "A" THEN

 SET total_time = route_a_base_time + delay_time

 PRINT "Total expected travel time: " + total_time + " minutes "

 //Computational thinking redirect pattern

 IF total_time > 50 THEN

 PRINT "ALERT: Route A is heavily delayed, Switch to Route B to
 save time."

 ENDIF

ELSE

 PRINT "Total expected travel time on Route B: " +

 route_b_base_time_ + " minutes. "

ENDIF

END